Basic Vulnerability Assessment for a Small Business Network

Submitted By – Nelakurthi Shiva

Problem Statement

To simulate a real-world vulnerability assessment for a small business IT infrastructure. Identify security gaps, prioritize risks, and provide mitigation strategies to improve the overall security posture.

Introduction

Vulnerability assessment is a systematic process used to identify, analyze, and mitigate security weaknesses in a system or network. With the increasing reliance on digital systems, identifying vulnerabilities has become essential to prevent cyber attacks.

In this project, a simulated environment was created using virtual machines to perform a vulnerability assessment on a target system. The assessment focuses on identifying open ports, running services, and associated vulnerabilities.

Objective

The objective of this project is:

- To identify open ports and services running on a system
- To detect vulnerabilities in the system
- To analyze risks associated with vulnerabilities
- To provide mitigation strategies to improve security

Tools Used

- Kali Linux
- Nmap
- VMware
- Metasploitable2

Lab Setup

A virtual lab environment was created using VMware.

The setup includes:

- Kali Linux as the attacker machine
- Metasploitable2 as the target machine

Both systems were configured within the same network to enable communication.

Methodology

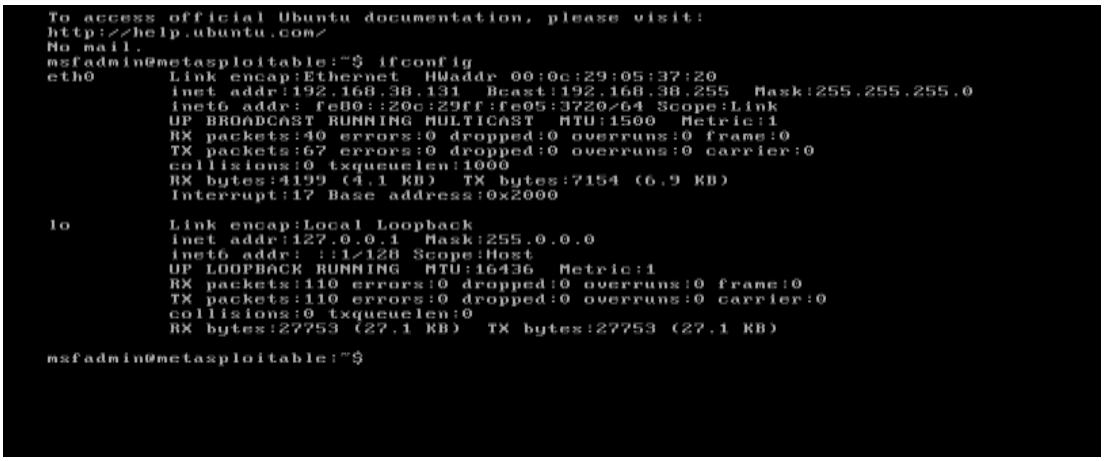
The following steps were followed:

- Setup of virtual lab environment
- Identification of target IP address
- Verification of connectivity using ping
- Network scanning using Nmap
- Identification of open ports and services
- Analysis of vulnerabilities
- Risk classification
- Recommendation of mitigation strategies

IP Address Identification

The IP address of the target machine was identified using the following command: ***ifconfig***

The identified IP address is: **192.168.38.131**



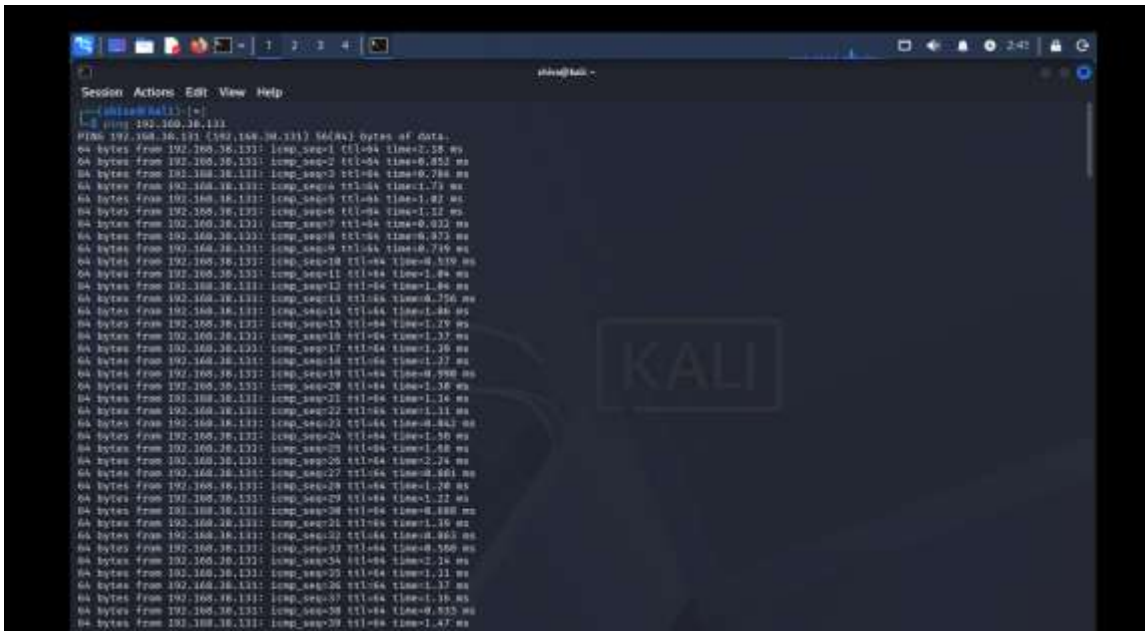
```
To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:05:37:20
          inet addr:192.168.38.131  Bcast:192.168.38.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe05:3720/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:40 errors:0 dropped:0 overruns:0 frame:0
          TX packets:67 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:4199 (4.1 KB)  TX bytes:7154 (6.9 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:110 errors:0 dropped:0 overruns:0 frame:0
          TX packets:110 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:27753 (27.1 KB)  TX bytes:27753 (27.1 KB)

msfadmin@metasploitable:~$
```

Figure 1: IP Address Identification using ifconfig

Connectivity Verification

Connectivity was verified using: **ping 192.168.38.131**

Figure 2: Connectivity Verification using Ping



Network Scanning

Network scanning was performed using Nmap:

nmap -sV 192.168.38.131

nmap -A 192.168.38.131

The scan identified multiple open ports and services running on the system.

Figure 3: Nmap Scan Results (-sV)

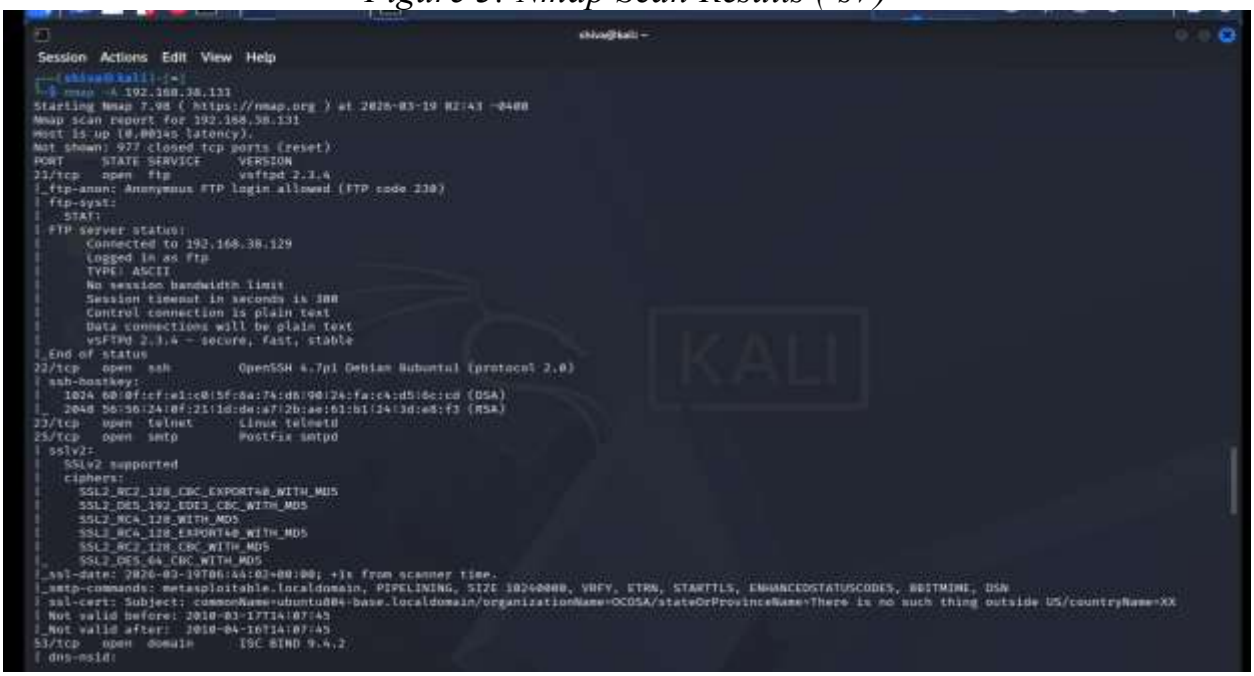


Figure 4: Detailed Nmap Scan Output (-A) (Part 1)

```
Session  Actions  Edit  View  Help
| dns-nsid:
|_  bind.version: 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_http-title: Metasploitable2 - Linux
111/tcp  open  rpcbind     2 (RPC #100000)
| rpcinfo:
|   program version    port/proto  service
|   100000  2            111/tcp   rpcbind
|   100000  2            111/udp   rpcbind
|   100003  2,3,4       2049/tcp   nfs
|   100003  2,3,4       2049/udp   nfs
|   100005  1,2,3      49413/tcp   mountd
|   100005  1,2,3      59212/udp   mountd
|   100021  1,3,4      36757/udp   nlockmgr
|   100021  1,3,4      56136/tcp   nlockmgr
|   100024  1          34610/udp   status
|_  100024  1          54932/tcp   status
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login       OpenBSD or Solaris rlogind
514/tcp  open  tcpwrapped
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
| mysql-info:
|   Protocol: 10
|   Version: 5.0.51a-3ubuntu5
|   Thread ID: 9
|   Capabilities flags: 43564
|   Some Capabilities: Speaks41ProtocolNew, Support41Auth, ConnectWithDatabase, LongColumnFlag, SupportsTransactions, SwitchToSSLAfterHandshake, SupportsCom
pression
|   Status: Autocommit
|_  Salt: Mv,u-px+d;!Qf0ZUy$Cd
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
|_ssl-date: 2026-03-19T06:46:02+00:00; +1s from scanner time.
| ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
| Not valid before: 2010-03-17T14:07:45
|_Not valid after:  2010-04-16T14:07:45
5900/tcp open  vnc         VNC (protocol 3.3)
```

Figure 5: Detailed Nmap Scan Output (-A) (Part 2)

```
Session  Actions  Edit  View  Help
5900/tcp open  vnc         VNC (protocol 3.3)
| vnc-info:
|   Protocol version: 3.3
|   Security types:
|_    VNC Authentication (2)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
| irc-info:
|   users: 1
|   servers: 1
|   lusers: 1
|   lservers: 0
|   server: irc.Metasploitable.LAN
|   version: Unreal3.2.8.1. irc.Metasploitable.LAN
|   uptime: 0 days, 0:09:41
|   source ident: nmap
|   source host: 2C310738.E3248194.FFFA6049.IP
|_  error: Closing Link: dhordimjd[192.168.38.129] (Quit: dhordimjd)
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
|_ajp-methods: Failed to get a valid response for the OPTION request
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
|_http-server-header: Apache-Coyote/1.1
|_http-title: Apache Tomcat/5.5
|_http-favicon: Apache Tomcat
MAC Address: 00:0C:29:05:37:20 (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)
|_clock-skew: mean: 1h00m01s, deviation: 2h00m00s, median: 0s
|_smb2-time: Protocol negotiation failed (SMB2)
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
| smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
```

Figure 6: Detailed Nmap Scan Output(-A) (Part 3)

```
Host script results:
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)
|_clock-skew: mean: 1h00m01s, deviation: 2h00m00s, median: 0s
|_smb2-time: Protocol negotiation failed (SMB2)
|_nbstat: NetBIOS name: METASPLOITABLE, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
| smb-os-discovery:
|   OS: Unix (Samba 3.0.20-Debian)
|   Computer name: metasploitable
|   NetBIOS computer name:
|   Domain name: localdomain
|   FQDN: metasploitable.localdomain
|_  System time: 2026-03-19T02:43:55-04:00

TRACEROUTE
HOP RTT     ADDRESS
1   1.41 ms 192.168.38.131

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 22.49 seconds

┌──(shiva㉿kali)-[~]
└─$
```

Scan Results

The following ports and services were identified:

Port	Service	Version
21	FTP	vsftpd 2.3.4
22	SSH	OpenSSH 4.7p1
23	Telnet	Linux telnetd
25	SMTP	Postfix
53	DNS	ISC BIND 9.4.2
80	HTTP	Apache 2.2.8
139/445	SMB	Samba 3.x
3306	MySQL	MySQL5.0
5432	PostgreSQL	PostgreSQL DB
8180	HTTP	Apache Tomcat

Vulnerabilities Identified

1. FTP (Port 21) – vsftpd 2.3.4

- Anonymous login is allowed
- Known backdoor vulnerability exists
- **Risk:** Unauthorized file access and potential remote exploitation

2. Telnet (Port 23)

- Uses unencrypted communication protocol
- Credentials transmitted in plain text
- **Risk:** Attackers can intercept login credentials (MITM attack)

3. HTTP (Port 80) – Apache 2.2.8

- Outdated web server version
- Vulnerable to multiple known exploits
- **Risk:** Web server compromise, data leakage

4. SMB (Ports 139/445) – Samba 3.x

- Message signing disabled
- Known vulnerabilities in older Samba versions
- **Risk:** Unauthorized access, lateral movement in network

5. MySQL (Port 3306)

- Running outdated version (MySQL 5.x)
- Weak authentication mechanisms possible
- **Risk:** Database compromise and data theft

6. PostgreSQL (Port 5432)

- Old version detected
- Default configurations may be insecure
- **Risk:** Sensitive data exposure

7. Apache Tomcat (Port 8180)

- Outdated Tomcat server
- Default credentials or misconfigurations possible
- **Risk:** Remote code execution and server compromise

9. SSH (Port 22) – OpenSSH

- Older version
- **Risk:** Potential brute-force attacks or vulnerabilities

Risk Analysis

Severity	Vulnerabilities
High	FTP, Telnet, SMB
Medium	Apache, Tomcat
Low	DNS, SSH

Mitigation Strategies

- Disable anonymous FTP access
- Replace Telnet with SSH
- Update Apache and Tomcat servers
- Enable SMB security configurations
- Update database services
- Use firewall rules to restrict access
- Perform regular security updates

Conclusion

The vulnerability assessment revealed multiple security issues in the system, primarily due to outdated services and insecure configurations. These vulnerabilities can be exploited by attackers to gain unauthorized access and compromise system security.

By implementing the recommended mitigation strategies, the system can be secured effectively.